

Exercises 1A

1. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function such that $L(f, P, [a, b]) = U(f, P, [a, b])$ for some partition P of $[a, b]$. Prove that f is the constant function on $[a, b]$.

Proof. Let $a = x_0, \dots, x_n = b$ denote the partition P of $[a, b]$. Since the lower and upper Riemann sums are equal, we have

$$\begin{aligned} \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f &= \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f. \\ \implies \sum_{j=1}^n (x_j - x_{j-1}) \left(\sup_{[x_{j-1}, x_j]} f - \inf_{[x_{j-1}, x_j]} f \right) &= 0 \end{aligned}$$

The length of each interval is strictly positive, and $\sup_{[x_{j-1}, x_j]} f \geq \inf_{[x_{j-1}, x_j]} f$ for each j , so the above sum consists of only non-negative terms. Thus, we have $(x_j - x_{j-1}) \left(\sup_{[x_{j-1}, x_j]} f - \inf_{[x_{j-1}, x_j]} f \right) = 0$ for each interval, and $\left(\sup_{[x_{j-1}, x_j]} f = \inf_{[x_{j-1}, x_j]} f \right)$. Thus, f is constant on every interval $[x_{j-1}, x_j]$.

$f(x) = f(a)$ for every $x \in [x_1, x_2]$. Suppose f is $f(a)$ on an interval $[x_{j-1}, x_j]$. Since f is also constant on the interval $[x_j, x_{j+1}]$, $f(x) = f(x_j) = f(a)$ for all $x \in [x_j, x_{j+1}]$. By mathematical induction, $f(x) = f(a)$ on every interval $[x_j, x_{j+1}]$. Therefore, f is constant on the entire domain. $\square$

2. Suppose that $a \leq s < t \leq b$. Define $f : [a, b] \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } s < x < t, \\ 0 & \text{otherwise.} \end{cases}$$

Prove that f is Riemann integrable on $[a, b]$ and that $\int_a^b f = t - s$.

Proof. Consider the partition P of $[a, b]$ into $n \in \mathbb{N}$ intervals of equal length given by $x_i = a + i \frac{b-a}{n}$. Notice that

$$L(f, P, [a, b]) = \sum_{i=1}^n d \inf_{[x_{i-1}, x_i]} f \geq \left(\left\lfloor \frac{n(t-s)}{b-a} \right\rfloor - 2 \right) \frac{b-a}{n},$$

and similarly

$$U(f, P, [a, b]) = \sum_{i=1}^n d \sup_{[x_{i-1}, x_i]} f \leq \left\lfloor \frac{n(t-s)}{b-a} \right\rfloor \frac{b-a}{n}.$$

Then,

$$L(f, [a, b]) \geq \sup_{n \in \mathbb{N}} \left(\left\lfloor \frac{n(t-s)}{b-a} \right\rfloor - 2 \right) \frac{b-a}{n} = t - s = \inf_{n \in \mathbb{N}} \left(\left\lfloor \frac{n(t-s)}{b-a} \right\rfloor \right) \frac{b-a}{n} \geq U(f, [a, b]).$$

But, since $L(f, [a, b]) \leq U(f, [a, b])$ we have $L(f, [a, b]) = U(f, [a, b]) = t - s$, and hence $\int_a^b f = t - s$. $\square$

3. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function. Prove that f is Riemann integrable if and only if for each $\epsilon > 0$ there exists a partition P of $[a, b]$ such that

$$U(f, P, [a, b]) - L(f, P, [a, b]) < \epsilon.$$

Proof.

($\implies$) Suppose f is Riemann integrable, and let $I := \int_a^b f$. Let $\epsilon > 0$. Since $I = \sup_P L(f, P, [a, b]) = \inf_P U(f, P, [a, b])$, there is a partition P_1 so that $L(f, P_1, [a, b]) > I - \frac{\epsilon}{2}$ and a partition P_2 so that $U(f, P_2, [a, b]) < I + \frac{\epsilon}{2}$. Let P be the partition obtained by taking the union of P_1 and P_2 . Since P is a finer partition than both P_1 and P_2 , we have

$$L(f, P, [a, b]) \geq L(f, P_1, [a, b]) > I - \epsilon/2$$

and

$$U(f, P, [a, b]) \leq U(f, P_2, [a, b]) < I + \epsilon/2.$$

Hence, $|L(f, P, [a, b]) - I| < \epsilon/2$ and $|U(f, P, [a, b]) - I| < \epsilon/2$. By the triangle inequality, we have

$$\begin{aligned} U(f, P, [a, b]) - L(f, P, [a, b]) &= |U(f, P, [a, b]) - L(f, P, [a, b])| \\ &= |U(f, P, [a, b]) - I + I - L(f, P, [a, b])| \\ &\leq |U(f, P, [a, b]) - I| + |I - L(f, P, [a, b])| \\ &< 2(\epsilon/2) = \epsilon, \end{aligned}$$

as desired.

($\impliedby$) Suppose that f is not Riemann integrable. Let $L := L(f, [a, b])$ and $U := U(f, [a, b])$ be the lower and upper Riemann integrals, where $L \neq U$. Since $L(f, P, [a, b]) \leq U(f, P, [a, b])$ for any partition, $L \leq U$. Let $\epsilon := \frac{U-L}{2}$. Then, for any partition P of $[a, b]$, we have

$$\begin{aligned} U(f, P, [a, b]) - L(f, P, [a, b]) &\geq \inf_P U(f, P, [a, b]) - \sup_P L(f, P, [a, b]) \\ &= U - L \\ &> \epsilon. \end{aligned}$$

□

4. Suppose $f, g : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable. Prove that $f + g$ is Riemann integrable and that $\int_a^b (f + g) = \int_a^b f + \int_a^b g$.

Proof. Recall that the infimum and supremum distribute over the minkowski sums of sets. Then, we have

$$\begin{aligned}
L(f + g, [a, b]) &= \sup_P L(f + g, P, [a, b]) \\
&= \sup_P \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f + g \\
&= \sup_P \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f + \sup_P \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} g \\
&= \int_a^b f + \int_a^b g \\
&= \inf_P \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f + \inf_P \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} g \\
&= \inf_P \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f + g \\
&= \inf_P U(f + g, P, [a, b]) \\
&= U(f + g, [a, b])
\end{aligned}$$

Hence, $L(f + g, [a, b]) = U(f + g, [a, b]) = \int_a^b f + \int_a^b g$. □

5. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Prove that $-f$ is Riemann integrable and that $\int_a^b (-f) = -\int_a^b f$.

Proof.

$$\begin{aligned}
L(-f, [a, b]) &= \sup_P \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} (-f) \\
&= -\inf_P \sum_{j=1}^n (x_j - x_{j-1}) \left(-\sup_{[x_{j-1}, x_j]} (f) \right) \\
&= -\inf_P \sum_{j=1}^n (x_j - x_{j-1}) \left(\sup_{[x_{j-1}, x_j]} (f) \right) \\
&= -\int_a^b f \\
&= -\sup_P \sum_{j=1}^n (x_j - x_{j-1}) \left(\inf_{[x_{j-1}, x_j]} (f) \right) \\
&= \inf_P \sum_{j=1}^n (x_j - x_{j-1}) \left(-\sup_{[x_{j-1}, x_j]} (-f) \right) = U(-f, [a, b])
\end{aligned}$$

□

6. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Suppose $g : [a, b] \rightarrow \mathbb{R}$ is a function such that $g(x) = f(x)$ for all except finitely many $x \in [a, b]$. Prove that g is Riemann integrable on $[a, b]$ and

$$\int_a^b g = \int_a^b f.$$

Proof. Let $h := g(x) - f(x)$ be a function defined on $[a, b]$. Since $g = f + h$, and f is Riemann integrable, it suffices to show that h is Riemann integrable. Let $x_1, \dots, x_n$ be the collection of all finitely many points so that $g(x_i) \neq f(x_i)$ for all i , given in increasing order. Moreover, let $C := \sup_{i=1, \dots, n} h(x_i) - \inf_{i=1, \dots, n} h(x_i)$.

Let $\epsilon > 0$, and $N := \lceil \frac{b-a}{\epsilon} \rceil$. Then, let $P_1 := \{a = y_0, y_1, \dots, y_N = b\}$ where $y_i = \min\{a + i\epsilon, b\}$, and let $P_2 := \{a, x_1, \dots, x_n, b\}$. Consider the finer partition $P = P_1 \cup P_2$, whose elements shall be indexed $a = z_0, z_1, z_2, \dots, z_M = b$. Then, we have

$$\begin{aligned} U(h, P, [a, b]) - L(h, P, [a, b]) &= \sum_{i=1}^M (z_i - z_{i-1}) \sup_{[z_{i-1}, z_i]} h - \sum_{i=1}^M (z_i - z_{i-1}) \inf_{[z_{i-1}, z_i]} h \\ &\leq \sum_{i=1}^M (z_i - z_{i-1}) \left| \sup_{[z_{i-1}, z_i]} h - \inf_{[z_{i-1}, z_i]} h \right| \\ &\leq \epsilon \sum_{i=1}^M \left| \sup_{[z_{i-1}, z_i]} h - \inf_{[z_{i-1}, z_i]} h \right| \end{aligned}$$

Notice that $\sup h$ is identically 0 on every interval $[z_{i-1}, z_i]$ that does not contain some x_j . Let $m_1, \dots, m_n$ be the indices so that $z_{m_j} = x_j$ for every $1 \leq j \leq n$. Then,

$$\begin{aligned} \epsilon \sum_{i=1}^M \left| \sup_{[z_{i-1}, z_i]} h - \inf_{[z_{i-1}, z_i]} h \right| &= \epsilon \sum_{i=1}^n \left| \sup_{[z_{m_i-1}, z_{m_i}]} h - \inf_{[z_{m_i-1}, z_{m_i}]} h \right| \\ &\leq \epsilon \cdot (nC) \end{aligned}$$

Since nC is constant and ϵ was arbitrary, we have found a partition so that $U(h, P, [a, b]) - L(h, P, [a, b]) \leq \epsilon$ for each ϵ . Hence, h is Riemann integrable, and consequently, so is g . $\square$

7. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function. For $n \in \mathbb{N}$, let P_n denote the partition that divides $[a, b]$ into 2^n intervals of equal size. Prove that

$$L(f, [a, b]) = \lim_{n \rightarrow \infty} L(f, P_n, [a, b]) \text{ and } U(f, [a, b]) = \lim_{n \rightarrow \infty} U(f, P_n, [a, b])$$

Proof. We prove the statement for the lower integral. Let $\epsilon > 0$. Since $L(f, [a, b]) = \sup_P L(f, P, [a, b])$, there is some partition P so that $|L(f, [a, b]) - L(f, P, [a, b])| < \epsilon$. Let $a = x_0, x_1, \dots, x_m = b$ denote the partition of P . Let $\delta = \min\{(x_i - x_{i-1}) : 1 \leq i \leq m\} \cup \{\epsilon\}$. Let $N \in \mathbb{N}$ be so that $\frac{b-a}{N} < \delta$. Then, where $n > N$, consider the partition $Q = P \cup P_n$.

Since Q is a refinement of P , we have $L(f, P, [a, b]) \leq L(f, Q, [a, b]) \leq L(f, [a, b])$. Moreover, since $L(f, [a, b]) - L(f, P, [a, b]) < \epsilon$, we have $L(f, Q, [a, b]) - L(f, P, [a, b]) < \epsilon$.

Let $M > 0$ be so that $|f(x)| < M$ for all $x \in [a, b]$ (since f is bounded). P_n consists of intervals of size at most δ . Q is a refinement of P_n obtained by adjoining the m points of P , each of which splits one of the δ -sized intervals. Each of these insertions contributes a difference of at most δM from the lower sum of $L(f, P_n, [a, b])$. Thus, altogether, we have $|L(f, Q, [a, b]) - L(f, P_n, [a, b])| \leq m\delta M$. Thus,

$$\begin{aligned} & |L(f, P_n, [a, b]) - L(f, [a, b])| \\ &= |L(f, P_n, [a, b]) - L(f, Q, [a, b]) + L(f, Q, [a, b]) - L(f, P, [a, b]) + L(f, P, [a, b]) - L(f, [a, b])| \\ &\leq |L(f, P_n, [a, b]) - L(f, Q, [a, b])| + |L(f, Q, [a, b]) - L(f, P, [a, b])| + |L(f, P, [a, b]) - L(f, [a, b])| \\ &\leq m\delta M + \epsilon + \epsilon \\ &\leq \epsilon(mM + 2). \end{aligned}$$

Since $mM + 2$ is constant and ϵ was arbitrary, we have found N so that when $n > N$, we have $L(f, P_n, [a, b])$ is arbitrarily close to $L(f, [a, b])$. Hence, $L(f, [a, b]) = \lim_{n \rightarrow \infty} L(f, P_n, [a, b])$. $\square$